Romania
Competition Council

The Romanian Competition Council ('RCC') must manage increasingly large volumes of data that cannot be stored and queried exclusively using traditional RDBMS (relational database management systems). On the other hand, in recent years, there was a clear trend to integrate more and more intelligent processes into applications, while functionality remains at the basic requirement level for ease of use. Also, since the abundance of sources implies a lot of disparate information, apparently unrelated to each other, predictive analyses implemented through Big Data technologies allow RCC to identify connections and patterns of interest, the identification of which would be impossible in the absence of these tools.

The development of the main component of our digital investigative system, the Big Data Platform, finalized at the end of 2021. This component allows the specialized staff of the RCC to access/use data for the initiation and development of cases and market studies based on 5 analysis modules: cartel screening, bid rigging detection, sector inquiries, mergers and structural and commercial links between enterprises.

An important aspect of our digital investigative processes is access to correct and complete data, their processing and interpretation. For these reasons, data acquisition is an important process and allows the downstream components of data processing to perform their functions correctly.

In the data acquisition process, the RCC platform administrators focus on several important aspects:

- picking up frequent data changes to ensure that the data destination is in sync with the content source;
- storing the data as it is retrieved; any data operation (normalization, cleaning, enhancement) is performed later in the loading/storage process;
- obtaining all the necessary metadata from the content source;
- allowing for the possibility to retrieve files stored in the content source.

The main data processes in the RCC BD platform are presented in the following diagram.

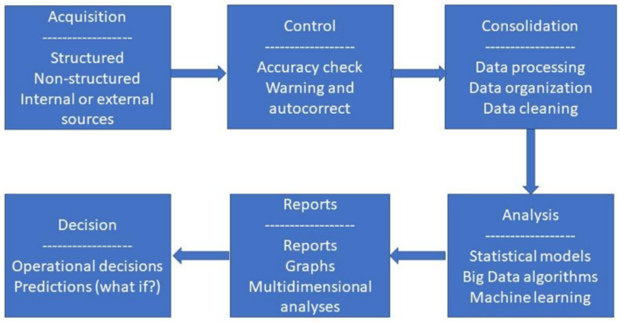

Figure I: data processes

I. Acquisition

The acquisition process consists of data imports from specific sources of internal or external data, representing transactional, operational data, content, references or system-generated data. Data is stored in relational databases, distributed file systems and NoSQL databases.

The external data which is part of our online architecture comes from the National Trade Registry, the tax authority, the electronic public procurement platform, the National Council for Solving Complaints in public procurement and the Ministry of Justice.

Apart from the external sources, the platform also utilizes most of the data RCC already stores since it automated internal workflows and digitized physical archives. The daily press report that RCC receives as part of an external contract is also uploaded to the platform. In addition, RCC already had digital databases which harbor important data, such as the Price Monitor app for fuels and food products and the State Aid Registry.

In the future, if deemed appropriate, an avenue that could be explored is to upload case documents into private and secure workspaces of the BD platform, where more advanced operations could be performed, such as text analytics.

II. Control and cleaning

A large portion of the structured data is imported from external sources. Until now, the digitalization process of Romanian public administration was hindered by a low level of interoperability of public administration IT systems. The existing systems are generally fragmented, being designed and developed in isolation by different state institutions, outside a coherent national framework. Consequently, the external databases providing input to our platform were built without having in hindsight the need to standardize variables. The structured datasets can contain errors, especially if the original data was input by humans and the content was not checked by automatic mechanisms.

Defining an appropriate data structure that connects company ownership and financial data, public procurement datasets and qualitative or processed unstructured data – and makes filtering across these possible – requires a very advanced understanding of each processed dataset. Therefore, during the implementation of the platform, the project team had to analyze the datasets and identify possibilities for designing an optimal structure for our platform, as well as potential errors that might arise when trying to match datasets.

For example, one company can appear under several differently spelled names: short, long, misspelled, etc. Since one of the main functionalities of the BD platform is centering all collected information around companies and their relevant markets, the possibility to match sets of data related to a specific company is an essential part of the database-building process. The companies registered in Romania have a unique numerical identifier allocated by the National Trade Register, therefore a deterministic matching was made using this unique ID to overcome issues related to companies appearing under different names in different datasets.

For certain aspects, code was written to automate the cleaning process. For example, the unique numerical identifiers needed to be standardized since some included the country code (RO) and some did not. Also, the line feed was eliminated from CSV files in which the character fields contained the LF (line feed) character in order to correctly navigate the structure of the received file.

Another matching issue with our imported datasets is the lack of unique identifiers for natural persons who are shareholders or administrators of a company.

Even if the original databases, belonging to the institutions from which we acquire data based on collaboration protocols, include unique identifiers for persons (Personal Numerical Code), compliance with the GDPR policy has allowed us access only to the name, date of birth, city and county of birth in free text format.

The automatic data acquisition processes do not contain matching algorithms for natural persons. Therefore, during specific case investigations, if there are multiple results in a query, RCC experts have to establish if several records belong to the same person and define a 'same person' relationship between the respective records, which may be time-consuming and defeats the purpose of the platform.

The GDPR issue came up earlier as well, at the setup of the platform, when acquiring entire structured databases from other public authorities raised GDPR concerns. RCC was already using an interoperability platform that allowed it to interrogate databases of other institutions. The inquiries were however specific to a particular company, it did not involve large data sets. The Big Data platform, on the other hand, needs access to entire sets of data to screen and issue red flags. To alleviate GDPR concerns as to why access to privileged information of so many companies is needed in one go, the RCC BD team had to carefully justify the process and explain the legal underpinnings. RCC has currently made steps toward amending national legislation in order to overcome these hurdles.

III. Conclusion

Maintaining and improving data sets is an ongoing exercise for which long-term commitment is needed and resources need to be allocated. Nevertheless, one of the biggest and immediate added values of the RCC BD platform is the quick access to integrated administrative datasets.